

Principal Component Analysis(PCA)

PCA – The Real-World Power of Eigenvectors

- Principal Component Analysis, or PCA, is a technique that reduces complex data into fewer dimensions while keeping as much important variation as possible.
- It's like summarizing a whole book into a few powerful sentences without losing the meaning.

Role of Eigenvectors & Eigenvalues in PCA

- **Eigenvectors → Directions of Maximum Variance**

They point to the new coordinate axes (principal components) that best capture the spread of the data.

- **Eigenvalues → Amount of Variance Captured**

They tell us how much information (variance) is preserved along each eigenvector.

- **Ranking by Eigenvalues**

The larger the eigenvalue, the more important that direction is.

- **Dimensionality Reduction**

By keeping only the top eigenvectors (with largest eigenvalues), we reduce features while still preserving most of the information.

Housing Dataset						
HouseID	Area_sqft	Bedrooms	Age_years	Dist_to_Center_km	Crime Index	Price_kUSD
H100	850	2	10	12.5	3.2	85
H101	1200	3	5	8	2.5	140
H102	950	2	20	14	4	95
H103	1500	4	2	6.5	2	200
H104	2000	4	15	18	4.5	320
H105	1100	3	8	9.5	3	130
H106	780	1	30	20	4.8	70
H107	1600	4	1	5	1.8	280
H108	1300	3	12	10	2.9	150
H109	1750	4	3	7	2.2	260
H110	900	2	25	16	4.1	90
H111	1400	3	7	9	2.7	175

What we are going to do?

- Often in real life, we have many features — like in housing data: size, bedrooms, age, price, location. Too many dimensions make patterns hard to see.
- Principal Component Analysis (PCA) helps us **summarize the data** by creating new axes (principal components) that capture the most important variation.
- The trick: these new axes are built from **eigenvectors** of the data's **covariance matrix**, and the eigenvalues tell us how much information each axis captures.
- So instead of staring at 6 messy features, we can look at just 2 or 3 principal components that explain most of the story.

PCA Case Study: Reducing 2D Data to 1D

X	Y
2	3
3	5
4	4
5	7
6	8

- Mean X (μ_x): $\frac{2+3+4+5+6}{5} = 4$
- Mean Y (μ_y): $\frac{3+5+4+7+8}{5} = 5.4$

Centered Data Table ($X_{centered} = X - \mu$):

Original (x, y)	Centered ($x - 4, y - 5.4$)
(2, 3)	(-2.0, -2.4)
(3, 5)	(-1.0, -0.4)
(4, 4)	(0.0, -1.4)
(5, 7)	(1.0, 1.6)
(6, 8)	(2.0, 2.6)

Step 1: Centering the Data

$$\text{Mean } X = \bar{x} = \frac{2+3+4+5+6}{5} = 4$$

$$\text{Mean } Y = \bar{y} = \frac{3+5+4+7+8}{5} = 5.4$$

PCA Case Study: Reducing 2D Data to 1D

X	Y
2	3
3	5
4	4
5	7
6	8

Step 2: The Covariance Matrix

$$\text{Var}(X) = \frac{\sum(x - \bar{x})^2}{n-1} = 2.5$$

$$\text{Var}(Y) = \frac{\sum(y - \bar{y})^2}{n-1} = 4.3$$

$$\text{Cov}(X, Y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n-1} = 3$$

Covariance Matrix

$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$

$$= \begin{bmatrix} 2.5 & 3 \\ 3 & 4.3 \end{bmatrix}$$

PCA Case Study: Reducing 2D Data to 1D

X	Y
2	3
3	5
4	4
5	7
6	8

4. Step 3: Finding Eigenvalues (λ)

Solve the characteristic equation $\det(C - \lambda I) = 0$:

$$\det \begin{bmatrix} 2.5 - \lambda & 3.0 \\ 3.0 & 4.3 - \lambda \end{bmatrix} = (2.5 - \lambda)(4.3 - \lambda) - (3.0)(3.0) = 0$$

$$\lambda^2 - 6.8\lambda + 10.75 - 9 = 0 \implies \lambda^2 - 6.8\lambda + 1.75 = 0$$

Using the quadratic formula, we find:

- $\lambda_1 \approx 6.53$ (The amount of information along PC1)
- $\lambda_2 \approx 0.27$ (The noise along PC2)

2-D Example

X	Y
2	3
3	5
4	4
5	7
6	8

Explained Variance Ratio

It represents the proportion, or percentage, of the total variance of the original dataset that is captured by each new principal component.

$$EVR_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3 + \cdots + \lambda_n}$$

$$EVR_1 = \frac{6.53}{6.53 + 0.27} = 0.96 \text{ (96\%)}$$

$$EVR_2 = \frac{0.27}{6.53 + 0.27} = 0.0396 \text{ (4\%)}$$

$$PC1 \approx 0.6 * x + 0.8 * y$$

PCA Case Study: Reducing 2D Data to 1D

X	Y
2	3
3	5
4	4
5	7
6	8

Normalization

$$\sqrt{1^2 + 1.34^2} = 1.67$$

$$\frac{1}{1.67} = 0.6$$

$$\frac{1.34}{1.67} = 0.8$$

5. Step 4: Finding the Principal Component (Eigenvector)

For the largest eigenvalue $\lambda_1 = 6.53$, solve $(C - 6.53I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} 2.5 - 6.53 & 3.0 \\ 3.0 & 4.3 - 6.53 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -4.03 & 3.0 \\ 3.0 & -2.23 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $-4.03v_1 + 3v_2 = 0 \implies v_2 \approx 1.34v_1$. Normalizing the vector $\begin{bmatrix} 1 \\ 1.34 \end{bmatrix}$ gives our **First Principal Component**:

$$\mathbf{v}_1 \approx \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

PCA Case Study: Reducing 2D Data to 1D

6. Step 5: Final Projection (The "Output")

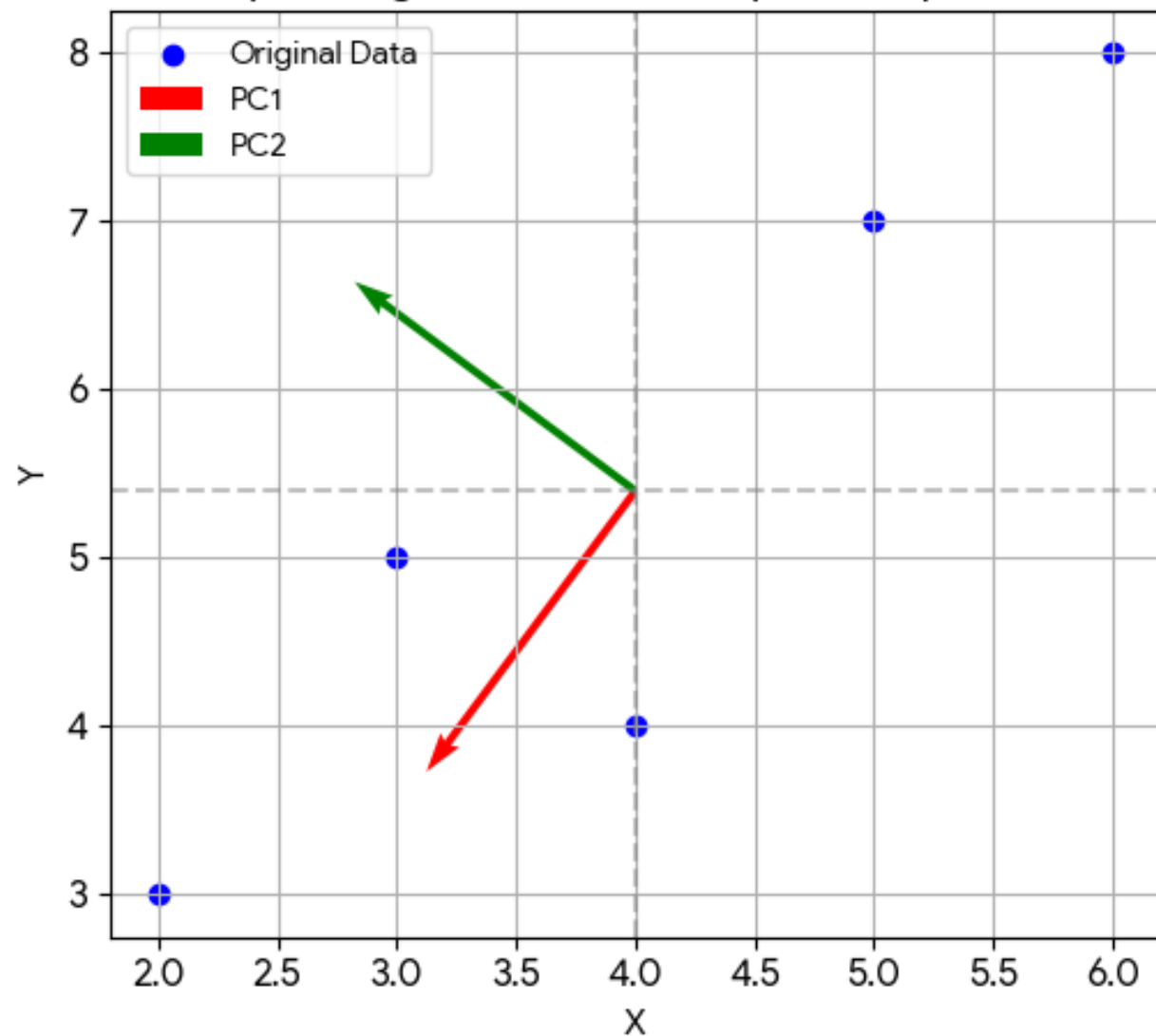
We can now represent each 2D point as a single 1D number (PC1 Score) by projecting the centered data onto our new basis vector $\mathbf{v}_1$:

$$\text{Score} = (x_{\text{centered}} \times 0.6) + (y_{\text{centered}} \times 0.8)$$

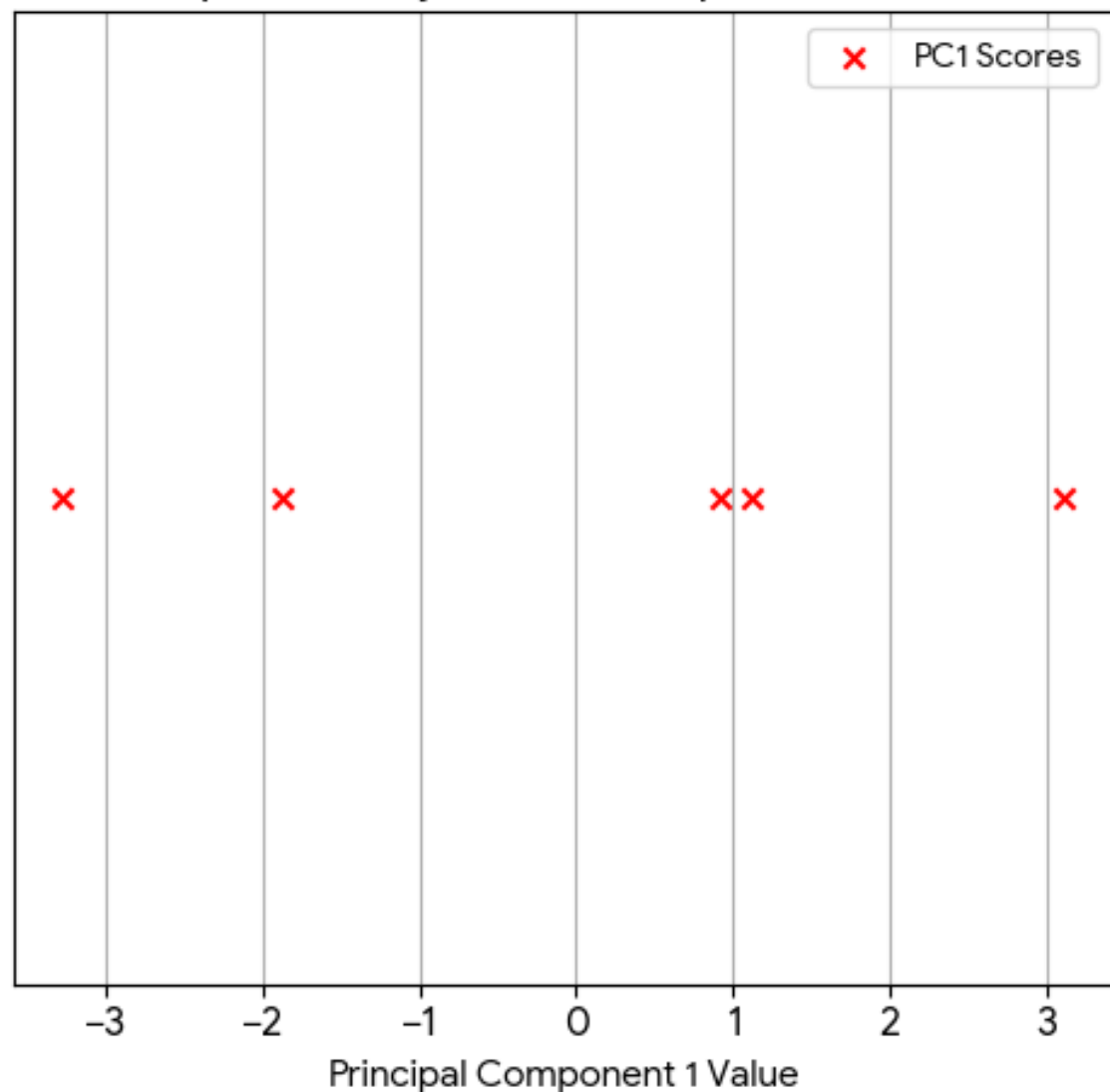
Centered (x, y)	PC1 Score (1D Projection)
$(-2.0, -2.4)$	$(-2.0 \cdot 0.6) + (-2.4 \cdot 0.8) = -3.12$
$(-1.0, -0.4)$	$(-1.0 \cdot 0.6) + (-0.4 \cdot 0.8) = -0.92$
$(0.0, -1.4)$	$(0.0 \cdot 0.6) + (-1.4 \cdot 0.8) = -1.12$
$(1.0, 1.6)$	$(1.0 \cdot 0.6) + (1.6 \cdot 0.8) = 1.88$
$(2.0, 2.6)$	$(2.0 \cdot 0.6) + (2.6 \cdot 0.8) = 3.28$

PCA Case Study: Reducing 2D Data to 1D

Step 1: Original Data & Principal Components



Step 2: 1D Projection (Compressed Data)



PCA is Feature Extraction not Feature Selection

Feature extraction transforms raw data into a new, reduced set of features, while feature selection chooses a subset of the original features and discards the rest

- **Process:**

Creates new features by combining, transforming, or reformatting existing ones

- **When to use:**

High-dimensional data where you need to capture underlying patterns or when original features are too complex.

Feature selection aims to remove irrelevant or redundant original features to simplify models and improve generalization.

In case of Housing, it may be $PC1 \approx 0.6 * \text{Size} + 0.4 * \text{Bedrooms} + 0.2 * \text{Age}$

$$\begin{bmatrix} Var(x_1) & Cov(x_1, x_2) \\ Cov(x_1, x_2) & Var(x_2) \end{bmatrix}$$

$$Cov(X, Y) = \frac{\sum (x_1 - \bar{x}_1)(x_2 - \bar{x}_2)}{n - 1}$$

Working Steps

- Find mean of x_1 and x_2 (i.e. $\bar{x}_1$ and $\bar{x}_2$)
- Construct columns $x_1 - \bar{x}_1$ and $x_2 - \bar{x}_2$
- Find Covariance Matrix
- Find Eigenvalues and normalized eigenvectors of covariance matrix
- Find PC_i 's score.

Example The dataset: 5 Houses $\times$ 3 Features

Three features: Rooms (count), Age (years), NearbyShops (count). We will compress this 3D data, 1D and see how much information survives.

House	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	Price (lakh)
	H1	1	20	2	20	H1	1	20	2	20	H1	1	20	2	20	
	H2	2	15	4	35	H2	2	15	4	35	H2	2	15	4	35	
	H3	3	10	6	50	H3	3	10	6	50	H3	3	10	6	50	
	H4	4	5	8	70	H4	4	5	8	70	H4	4	5	8	70	
	H5	5	2	10	90	H5	5	2	10	90	H5	5	2	10	90	
H1	1					20					2					20
H2	2					15					4					35
H3	3					10					6					50
H4	4					5					8					70
H5	5					2					10					90

Example The dataset: 5 Houses × 3 Features

Step 2: Mean center each column

$\text{mean(Rooms)} = \bar{x} = (1+2+3+4+5)/5 = \mathbf{3.0}$
 $\text{mean(Age)} = \bar{y} = (20+15+10+5+2)/5 = \mathbf{10.4}$
 $\text{mean(Shops)} = \bar{z} = (2+4+6+8+10)/5 = \mathbf{6.0}$

House	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$
	H1	-2	9.6	-4	H1	-2	9.6	-4	H1	-2	9.6	-4
	H2	-1	4.6	-2	H2	-1	4.6	-2	H2	-1	4.6	-2
	H3	0	-0.4	0	H3	0	-0.4	0	H3	0	-0.4	0
	H4	1	-5.4	2	H4	1	-5.4	2	H4	1	-5.4	2
	H5	2	-8.4	4	H5	2	-8.4	4	H5	2	-8.4	4
H1	-2			9.6			-4					
H2	-1			4.6			-2					
H3	0			-0.4			0					
H4	1			-5.4			2					
H5	2			-8.4			4					

House	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	House	Rooms x	Age (yrs) y	Shops z	Price (lakh)	Price (lakh)
	H1	1	20	2	20	H1	1	20	2	20	H1	1	20	2	20	
	H2	2	15	4	35	H2	2	15	4	35	H2	2	15	4	35	
	H3	3	10	6	50	H3	3	10	6	50	H3	3	10	6	50	
	H4	4	5	8	70	H4	4	5	8	70	H4	4	5	8	70	
	H5	5	2	10	90	H5	5	2	10	90	H5	5	2	10	90	
H1	1				20				2				20			
H2	2				15				4				35			
H3	3				10				6				50			
H4	4				5				8				70			
H5	5				2				10				90			

Example The dataset: 5 Houses × 3 Features

Step 3: Co-Variance

House	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
	$\begin{matrix} x & y & z \\ x - \bar{x} & y - \bar{y} & z - \bar{z} \\ (x - \bar{x})^2 & (y - \bar{y})^2 & (z - \bar{z})^2 \\ (x - \bar{x})(y - \bar{y}) & (x - \bar{x})(z - \bar{z}) & (y - \bar{y})(z - \bar{z}) \end{matrix}$									
H1	-2	9.6	-4	$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	
H2	-1	4.6	-2	$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	
H3	0	-0.4	0	$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	
H4	1	-5.4	2	$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	
H5	2	-8.4	4	$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	
				$\sum_{i=1}^5 (x - \bar{x})^2 = 10$	$\sum_{i=1}^5 (y - \bar{y})^2 = 2132$	$\sum_{i=1}^5 (z - \bar{z})^2 = 40$	$\sum_{i=1}^5 (x - \bar{x})(y - \bar{y}) = -46$	$\sum_{i=1}^5 (x - \bar{x})(z - \bar{z}) = -20$	$\sum_{i=1}^5 (y - \bar{y})(z - \bar{z}) = -92$	

Variance and Co-Variance Calculations

$$\text{Var}(x) = \frac{\sum(x - \bar{x})^2}{n-1} = \frac{10}{4} = 2.5$$

$$\text{Var}(y) = \frac{\sum(y - \bar{y})^2}{n-1} = \frac{213.2}{4} = 53.3$$

$$\text{Var}(z) = \frac{\sum(z - \bar{z})^2}{n-1} = \frac{40}{4} = 10$$

$$\text{Cov}(x, y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n-1} = -\frac{46.4}{4} = -11.5$$

$$\text{Cov}(x, z) = \frac{\sum(x - \bar{x})(z - \bar{z})}{n-1} = \frac{20}{4} = 5$$

$$\text{Cov}(y, z) = \frac{\sum(y - \bar{y})(z - \bar{z})}{n-1} = -\frac{92}{4} = -23$$

Co-Variance Matrix

$$\begin{bmatrix} \text{var}(x) & \text{CoVar}(x, y) & \text{CoVar}(x, z) \\ \text{covar}(x, y) & \text{var}(y) & \text{covar}(y, z) \\ \text{covar}(z, x) & \text{covar}(y, z) & \text{var}(z) \end{bmatrix} = \begin{bmatrix} 2.5 & -11.5 & 5 \\ -11.5 & 53.3 & -23 \\ 5 & -23 & 10 \end{bmatrix}$$

Eigen Values

$$\begin{vmatrix} 2.5 - \lambda & -11.5 & 5 \\ -11.5 & 53.3 - \lambda & -23 \\ 5 & -23 & 10 - \lambda \end{vmatrix} = 0$$

$$\lambda^3 - 65.8\lambda^2 + 5\lambda = 0$$

$$\lambda = 0, \quad 65.7, \quad 0.07$$

Eigen Vectors

$$\text{For } \lambda = 65.7, \quad [0.5, -2.3, 1]$$

$$\text{For } \lambda = 0.07, \quad [0.5, 0.54, 1]$$

$$\text{For } \lambda = 0, \quad [-2, 0, 1]$$

Explained Variance Ratio

$$EVR_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3 + \cdots + \lambda_n}$$

$$EVR_1 = \frac{65.7}{65.7 + 0.07 + 0} = 0.999$$

$$EVR_2 = \frac{0.07}{65.7 + 0.07 + 0} = 0.001$$

$$EVR_3 = \frac{0}{65.7 + 0.07 + 0} = 0$$

Eigenvectors: the new axis directions

After Normalization

PC1: $[0.5, -2.3, 1] \Rightarrow [0.2, -0.9, 0.39]$

PC2: $[0.5, 0.54, 1] \Rightarrow [0.4, 0.44, 0.81]$

PC3: $[-2, 0, 1] \Rightarrow [-0.89, 0, 0.45]$

Feature	PC1 ($\lambda=65.7$)	PC2 ($\lambda=0.07$)	PC3 ($\lambda=0$)
Rooms	0.2	0.4	-0.89
Age	-0.9	0.44	0
Shops	0.39	0.81	0.45

Reduce to 1D

$$\text{H1: } 0.2 \times (-2) + (-0.9) \times (9.6) + 0.39 \times (-4) = \mathbf{-10.6}$$

$$\text{H2: } 0.2 \times (-1) + (-0.9) \times (4.6) + 0.39 \times (-2) = \mathbf{-5.12}$$

$$\text{H3: } 0.2 \times (0) + (-0.9) \times (-0.4) + 0.39 \times (0) = \mathbf{0.36}$$

$$\text{H4: } 0.2 \times (1) + (-0.9) \times (-5.4) + 0.39 \times (2) = \mathbf{5.84}$$

$$\text{H5: } 0.2 \times (2) + (-0.9) \times (-8.4) + 0.39 \times (4) = \mathbf{9.52}$$

House	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$	House	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$
	H1	-2	9.6	-4	H1	-2	9.6	-4	H1	-2	9.6	-4
	H2	-1	4.6	-2	H2	-1	4.6	-2	H2	-1	4.6	-2
	H3	0	-0.4	0	H3	0	-0.4	0	H3	0	-0.4	0
	H4	1	-5.4	2	H4	1	-5.4	2	H4	1	-5.4	2
	H5	2	-8.4	4	H5	2	-8.4	4	H5	2	-8.4	4
H1		-2	9.6	-4								
H2		-1	4.6	-2								
H3		0	-0.4	0								
H4		1	-5.4	2								
H5		2	-8.4	4								

House	Rooms	Age	Shops	PC1 score (only)	Price
H1	1	20	2	-10.6	20
H2	2	15	4	-5.12	35
H3	3	10	6	0.36	50
H4	4	5	8	5.84	70
H5	5	2	10	9.52	90

Most negative (H1 = -10.6)

→ cheapest (20 lakh).

Most positive (H5 = +9.52)

→ most expensive

(90 lakh).

Thank You!

